

ADK Streaming Proof of Concept (POC) Documentation

Link:

Overview

This document outlines a complete end-to-end proof of concept (POC) for using **Google's ADK Streaming** with **FastAPI** to build a real-time, bidirectional, low-latency communication agent. The agent is enhanced with **Google Search grounding** and supports text, voice, and video interactions.

1. Prerequisites

- Python 3.8+
 - Familiarity with terminal usage (Windows, macOS, Linux)
 - API Key from [Google AI Studio](#)
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2. Installation

Create and Activate a Virtual Environment

```
# Create
python -m venv .venv

# Activate
# macOS/Linux:
source .venv/bin/activate
# Windows CMD:
.venv\Scripts\activate.bat
# Windows PowerShell:
.venv\Scripts\Activate.ps1
```

Install ADK

```
pip install google-adk fastapi uvicorn python-dotenv
```

3. Project Structure

```
adk-streaming/  
├── app/  
│   ├── .env          # Gemini API Key  
│   ├── main.py       # FastAPI app  
│   ├── google_search_agent/  
│   │   ├── __init__.py  
│   │   └── agent.py   # Agent definition  
│   └── static/  
│       └── index.html # Client UI
```

4. Agent Definition

agent.py

```
from google.adk.agents import Agent  
from google.adk.tools import google_search  
  
root_agent = Agent(  
    name="basic_search_agent",  
    model="gemini-2.0-flash-live-001",  
    description="Agent to answer questions using Google Search.",  
    instruction="You are an expert researcher. You always stick to the facts.",  
    tools=[google_search]  
)
```

__init__.py

```
from . import agent
```

5. Environment Setup

.env

```
GOOGLE_API_KEY=YOUR_API_KEY_HERE  
GOOGLE_GENAI_USE_VERTEXAI=0
```

6. FastAPI Streaming Backend

main.py

(Refer to full code provided in the original quickstart. It includes:)

- API Key loading
- Agent session management
- Bidirectional WebSocket streaming
- Static file serving

Run the server:

```
cd app  
uvicorn main:app --reload
```

Access the app at: <http://localhost:8000>

7. Client Web App

index.html

- Simple UI with message input and streaming output
- WebSocket-based communication

- Auto reconnection logic
 - Live message updates with `turn_complete` detection
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8. Testing

Using `adk web`

```
cd app
adk web
```

Open browser and test prompts via:

- Text input
- Voice button
- Video input

Using Custom Web App

1. Start FastAPI server: `uvicorn main:app --reload`
 2. Open browser: `http://localhost:8000`
 3. Type a message and view streaming response
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9. Benefits of ADK Streaming

- Real-time bi-directional communication
 - Low latency and seamless updates
 - Multimodal interaction: voice, video, image (future-ready)
 - Extensible architecture with FastAPI and WebSockets
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10. Future Enhancements

- Add audio/video streaming using WebRTC/WebSockets
- Enhance UI with rich multimodal support

- Implement custom tools (ShellAgent, LongRunningTool, etc.)
 - Integrate with cloud deployment (GCP, AWS, etc.)
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11. Resources

- [Google ADK Docs](#)
 - [Google AI Studio](#)
 - [FastAPI Documentation](#)
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